Chem Lab CSE

Summer 2025

Instructions:

- ➔ You will have 50 minutes to complete your exam. At the end of this time period, you must stop working. Partial credit will be awarded, so attempt as many questions as you can!
- ➔ [REFERENCE MATERIALS]Each team is allowed 2 note sheets with information on both sides, it must be printed and laminated or in a sheet protector.
- ➔ Ties will be broken in accordance with the national Science Olympiad rules in this order:
  - ⧫ Tiebreaker Questions answered Correctly

Name(s): _____

Score: __/60

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Questions? Comments? Email me:
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Section A: MCQ [30 pts]

MCQs

1. Which of the following increases the rate of a chemical reaction?
   a) Decreasing temperature
   b) Decreasing reactant concentration
   c) Increasing surface area of a solid reactant
   d) Using an inert solvent

2. The rate law for the reaction A + B → C is rate = $k[A]^2[B]$. What happens to the rate if [A] is halved and [B] is doubled?
   a) Rate is unchanged
   b) Rate is halved
   c) Rate is quartered
   d) Rate is doubled

3. A reaction has units of rate = $mol \cdot L^{-1} \cdot s^{-1}$ and k = 0.50 $L \cdot mol^{-1} \cdot s^{-1}$. What is the order of the reaction?
   a) Zero
   b) First
   c) Second
   d) Third

4. What is the activation energy for a reaction if its rate doubles for every 10°C increase in temperature?
   a) ~25 kJ/mol
   b) ~50 kJ/mol
   c) ~100 kJ/mol
   d) ~200 kJ/mol

5. Which of the following affects the value of the rate constant (k)?
   a) Pressure
   b) Temperature
   c) Catalyst presence
   d) All of the above

6. In the decomposition of N_2O_5, the rate law is rate = $k[N_2O_5]$. Which graph yields a straight line?
   a) $[N_2O_5]$ vs. time
   b) $\ln[N_2O_5]$ vs. time
   c) $1/[N_2O_5]$ vs. time
   d) ln(rate) vs. time

7. Which compound will undergo the fastest SN1 reaction?
   a) CH3Br
   b) CH3CH2Br
   c) (CH3)3CBr
   d) CH3CH2CH2Br

8. Which of the following is **not** a valid elementary step in a reaction mechanism?
   a) $A \rightarrow B$
   b) $A + B \rightarrow C$
   c) $2A + B \rightarrow C + D$
   d) $A + B + C + D \rightarrow E$

9. What is the order of the reaction with the following data?

[A] (M)	Rate (M/s)
0.10	0.020
0.20	0.080
0.40	0.320

   a) 0
   b) 1
   c) 2
   d) 3

10. The half-life of a first-order reaction depends on:
    a) Initial concentration
    b) Temperature
    c) Catalyst
    d) Both b and c

11. Which of the following reactions will proceed fastest at room temperature?
    a) Tertiary alkyl bromide with H_2O
    b) Primary alkyl chloride with OH^-
    c) Vinyl bromide with NH_3
    d) Aryl chloride with H_2O

12. What is the mole ratio of H_2O to CO_2 in the combustion of propane (C_3H_8)?
    a) 2:3
    b) 3:4
    c) 4:3
    d) 5:3

13. Which of the following is **true** for an elementary reaction?
    a) The rate law must be determined experimentally
    b) The rate law can be written directly from the stoichiometry

c) It involves multiple steps
d) It has no transition state

14. Which of the following reacts fastest with NaOH in an SN2 mechanism?
    a) 1-bromopropane
    b) 2-bromopropane
    c) 2-bromo-2-methylpropane
    d) Allyl bromide

15. A first-order reaction goes from 0.4 M to 0.1 M in 60 s. What is its half-life?
    a) 15 s
    b) 20 s
    c) 30 s
    d) 60 s

16. Which of the following organic compounds is **least** reactive in electrophilic aromatic substitution?
    a) Toluene
    b) Nitrobenzene
    c) Phenol
    d) Aniline

17. The rate-determining step of a reaction:
    a) Always involves the reactant with the highest concentration
    b) Has the highest activation energy
    c) Is the fastest step
    d) Determines the overall equilibrium constant

18. What is the limiting reagent in the reaction of 4.0 g H_2 with 28.0 g N_2 to form NH_3?
    a) H_2
    b) N_2
    c) NH_3
    d) Cannot be determined

19. A reaction is zero order in A and first order in B. Doubling [B] does what to the rate?
    a) Doubles it
    b) Halves it
    c) No change
    d) Quadruples it

20. Which species is the best nucleophile in polar aprotic solvents?
    a) F^-
    b) Cl^-
    c) Br^-

d) I^-

21. What is the percent yield if the theoretical yield is 5.2 g and the actual yield is 3.9 g?
    a) 67%
    b) 75%
    c) 80%
    d) 85%

22. What is the rate law for a reaction with this mechanism?
    Step 1: A + B ⇌ C (fast equilibrium)
    Step 2: C → D (slow)
    a) rate = k[A][B]
    b) rate = k[C]
    c) rate = k[A]
    d) rate = k[B]

23. What is the theoretical mass of water produced by combusting 5.0 g of methane?
    a) 5.6 g
    b) 9.0 g
    c) 11.3 g
    d) 13.5 g

24. Which undergoes E1 elimination fastest?
    a) 1-butanol
    b) 2-butanol
    c) tert-butanol
    d) Methanol

25. What is the molecularity of the step: A + B → C?
    a) Unimolecular
    b) Bimolecular
    c) Termolecular
    d) Cannot be determined

26. Which best describes the transition state in a reaction?
    a) The reactants
    b) The products
    c) A high-energy intermediate
    d) The activated complex

27. Which is a valid unit for a third-order rate constant?
    a) s^{-1}
    b) $L \cdot mol^{-1} \cdot s^{-1}$
    c) $mol^{-2} \cdot L^{2} \cdot s^{-1}$

d) $L^2 \cdot mol^{-2} \cdot s^{-1}$

28. How many grams of $CaCl_2$ are needed to prepare 250 mL of a 0.500 M solution? (Molar mass = 110.98 g/mol)
    a) 13.8 g
    b) 27.7 g
    c) 55.5 g
    d) 110.0 g

29. What is the theoretical number of molecules in 5.0 moles of a substance?
    a) 3.01×10^{23}
    b) 6.02×10^{23}
    c) 1.20×10^{24}
    d) 3.01×10^{24}

30. Which statement is true about SN2 reactions?
    a) The rate is independent of the nucleophile
    b) A carbocation intermediate forms
    c) They occur in two steps
    d) The reaction rate increases with less steric hindrance

Section B: FRQs [30 pts]

1. **[5 pts]** For the following reaction:
   $2A + B \rightarrow C$
   The rate law is rate = $k[A][B]^2$.
   Initial concentrations: [A] = 0.2 M, [B] = 0.1 M, and k = 0.500 $L^2 \cdot mol^{-2} \cdot s^{-1}$.
   Calculate the initial rate.

2. **[4 pts]** A reaction is first-order with respect to A. It takes 300 s for [A] to decrease from 0.80 M to 0.40 M. What is the rate constant?

3. **[4 pts]** Write the complete reaction and balanced net ionic equation for the combustion of 2-propanol. Then determine the number of moles of O_2 required to fully combust 100 g of 2-propanol.

4. **[4 pts]** The reaction $2NO + Cl_2 \rightarrow 2NOCl$ has the following rate law:
   rate = $k[NO]^2[Cl_2]$. If the rate doubles when $[Cl_2]$ is doubled, but quadruples when [NO] is doubled, explain the implications for the mechanism.

5. **[4 pts]** Calculate the molar solubility of Ag_2CrO_4 in 0.2 M Na_2CrO_4. ($K_{sp} = 1.2 \times 10^{-12}$)

6. **[3 pts]** The activation energy of a reaction is 75 kJ/mol. Calculate the rate constant ratio (k_2/k_1) when temperature increases from 300 K to 320 K..

7. **[3 pts]** Determine the limiting reagent and theoretical yield in grams of benzyl alcohol produced from 12.0 g of benzyl chloride and 4.0 g of NaOH.

KEY: